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RESEARCH ARTICLE

A Blockchain-Based Secure Distributed Authentication Scheme for Internet of Vehicles

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ABSTRACT Authentication for the Internet of Vehicles (IoV) is essential for safe IoV connection. However, because of the peculiar structure of IoV, current authentication schemes heavily rely on Road Side Units (RSUs). An excessive number of RSUs will make it difficult to synchronize updates, which will make system authentication inefficient overall and prevent the reuse of authentication results. Based on this, this paper proposes a blockchain-based secure distributed authentication scheme for IoV. First, data processing and storage are decentralized to the edge of the network to reduce the communication delay and response time with the trusted authority (TA). Second, in the IoV authentication process, smart contract technology is used to achieve automatic triggering of the authentication process, and an optimized Practical Byzantine Fault Tolerant Consensus (PBFT) algorithm is designed to add the authentication information to the blockchain ledger so that the authentication results can be reused. Finally, this paper performs formal security verification and informal security analysis using and Real-Or-Random (ROR) oracle model and AVISPA tool. In addition, this paper performs a detailed evaluation of computation cost and communication cost and uses SUMO for simulation verification. The security analysis and performance evaluation show that the scheme meets the security requirements of the IoV while having less communication and computation cost.

INDEX TERMS Blockchain, distributed authentication, Internet of Vehicles, consensus.

I. INTRODUCTION

With the progress of science and technology, the Internet of Vehicles (IoV) is becoming more and more prominent in intelligent transportation systems [1]. Its core concept is to realize real-time information interaction between vehicles and vehicles, vehicles and Road Side Units (RSUs), and vehicles and cloud data centers, providing solid backing support for intelligent transportation. However, the IoV still face many thorny problems, especially in security, such as illegal data injection, location data spoofing, data replay, denial of service, data tampering, and repudiation, etc. [2], [3], and [4], which bring serious challenges to the healthy development of IoV. Therefore, access authentication is particularly important [5]. Although existing authentication schemes for IoV can guarantee the security of the system to a certain extent, these schemes usually rely on a large number of RSUs communicating frequently with Trusted Authority (TA) [6], which undoubtedly increases the complexity of the system. Meanwhile, due to the dynamic and distributed nature of the IoV, the authentication results often cannot be effectively reused.

In recent years, the rise of blockchain technology, with its decentralized, traceable, and tamper-proof characteristics, has provided a new solution for vehicle authentication [7]. Vehicles can be safely validated without depending on a centralized TA by utilizing blockchain technology. Simultaneously, the distributed architecture of the blockchain

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facilitates the safe exchange and reliable transmission of transaction data throughout the network, offering robust backing for the secure verification of automobiles. However, with the rapid growth of the total number of automobiles, more and more cars are connected to the network, and it has become inevitable that the number of devices connected to the IoV is rising [8]. In this situation, the IoV must operate in low latency mode. To solve this problem, some scholars have proposed the solution of using edge computing technology. Edge computing reduces the communication delay and load with the cloud by decentralizing data processing and storage to the edge of the network [9], which improves authentication efficiency and reduces the dependence on RSUs. In addition, edge computing enables fast processing and storage of local data, improving the responsiveness and flexibility of the system [10].

In this paper, we propose a distributed trusted authentication scheme based on blockchain to ensure the security of the connected the IoV while improving communication latency and increasing authentication efficiency. The main contributions of this paper are as follows.

1) A blockchain-based secure distributed network model for the IoV is built to reduce communication delays and loads with TA while ensuring security by decentralizing data processing and storage to the edge of the network.
2) Through smart contract technology, the authentication process is combined with the blockchain to realize the automation and intelligence of the authentication process. Meanwhile, to achieve the reuse of authentication results, the authentication information is added to the blockchain ledger through an optimized consensus algorithm, which effectively reduces the communication cost.
3) A comprehensive security analysis of the proposed scheme is carried out using the ROR oracle model and AVISPA tool. The computational cost and communication cost are also evaluated in detail and simulations are performed using sumo for verification, thus ensuring the reliability of the scheme.

This paper is organized as follows. Section I provides introductory work on the paper. Section II reviews the related work and Section III presents the preliminary work and the proposed system model. Section IV presents the details of the design program. Section V provides an informal and formal security discussion of the scheme. Section VI evaluates the performance of the scheme and compares it with related schemes. Section VII summarizes the paper and discusses future work.

II. RELATED WORK

The main architecture of traditional IoV consists mainly of vehicles, RUSs, and trusted authority. However, this architecture is susceptible to single-point-of-failure issues. As a result, researchers have begun exploring various ways to use blockchain technology to design authentication schemes for the Internet of Cars. For example, Grover [11] provides a detailed review of blockchain security solutions for the Internet of Cars and highlights the critical role of blockchain technology in identity scenarios. Using blockchain technology, Zheng et al. [12] used a secure access authentication scheme between vehicles and RUSs and constructed a traceable and decentralized IoV system. The system allows vehicles to communicate anonymously using pseudonyms in environments that are not fully trusted and effectively enables transparency in the authentication and announcement of vehicles through blockchain technology. Furthermore, Noh et al. [13] proposed a message authentication scheme using blockchain technology to achieve message anonymity and decentralization, where security is guaranteed, but at the same time it increases the computational cost. Vangala et al. [14] innovated the use of hybrid blockchain technology on mobile agricultural vehicles incorporating elliptic curve operations to significantly reduce computational and communication costs. Xue et al. [15], on the other hand, proposed a distributed authentication protocol using smart contracts that aims to provide roaming services in vehicular mobility networks. However, the protocol employs weak anonymity by encrypting identities and fails to give unlinkability and untraceability. Moreover, the protocol contains a non-clonable device using public cryptography, which may increase the computational burden. Although blockchain technology has been leveraged in all these approaches to enhance data security, it also limits the ability to process frequent transactions to some extent. Integrating Blockchain and Edge Computing is a Promising Paradigm [16], [17], the security and privacy benefits of blockchain and the need for increased scalability are complemented by the benefits of distributed implementation of scalability and the need for secure and efficient control of edge computing [18]. Shen et al. [19] implements mutual authentication between vehicles, edge nodes and cloud servers. Guo et al. [20], on the other hand, proposed a layered architecture including in vehicle network layer, block-chain edge layer, and blockchain network layer, which realizes fast authentication of vehicles and collaborative sharing among vehicle networks. However, the authenticity of vehicle identity is not guaranteed. To solve this problem, Yao et al. [21] proposed a distributed blockchain-assisted lightweight anonymous authentication BLA mechanism based on blockchain. This mechanism reduces the communication delay in the authentication process by reducing the interaction between the vehicle and the server and writes the authentication result to the block using the PBFT consensus algorithm. When the vehicle moves to the next service zone, there is no need to authenticate again. Wu et al. [22], on the other hand, proposed an efficient and secure distributed IoV device authentication mechanism based on the combination of blockchain and mobile edge computing computation.

In addition, consensus algorithms are key in distributed systems. Kim [23] conducted a comparative analysis of three

consensus algorithms, PoW, PoS, and PBFT, and found that PBFT has the highest promise as a consensus algorithm in vehicular networks. However, the communication complexity of PBFT increases with the number of nodes. When the number of nodes exceeds 100 [24], it can significantly impact the scalability of the blockchain. Therefore, in complex IoV with a large number of connected vehicles, there is a need to improve the communication efficiency of consensus algorithms. To address this problem, Kumar et al. [25] used the reputation mechanism of logistic regression computation to enhance the PBFT consensus process and proposed a fast and intelligent consensus mechanism algorithm R-PBFT. However, the algorithm does not take into account the effect of a large number of mobile nodes in the network on the consensus efficiency. Xu et al. [26] further improved the consensus efficiency of the system by grouping nodes based on a scoring system. However, this approach needs to address the communication cost. To address the problem of high latency and resource consumption, Zhou et al. [27] proposed a block-chain-assisted spectrum transaction framework based on the Dynamic Credit Aggregate Signature Byzantine Fault Tolerance (DCABFT) algorithm in conjunction with BLS aggregated signatures. The framework enables an efficient spectrum transaction by utilizing BLS aggregated signature technology while ensuring data security and trustworthiness.

III. PRELIMINARIES AND SYSTEM MODEL

This section discusses the related theoretical foundations, including blockchain technology as well as BLS aggregated signatures. It also introduces the blockchain-based secure distributed authentication model for IoV proposed in this scheme.

A. BLOCKCHAIN

Blockchain is one of the emerging technologies, which is essentially a decentralized distributed database maintained by all nodes. As shown in Figure 1, to construct blocks for a blockchain-based system, distributed nodes ensure that all nodes maintain the same copy of the blockchain through consensus.

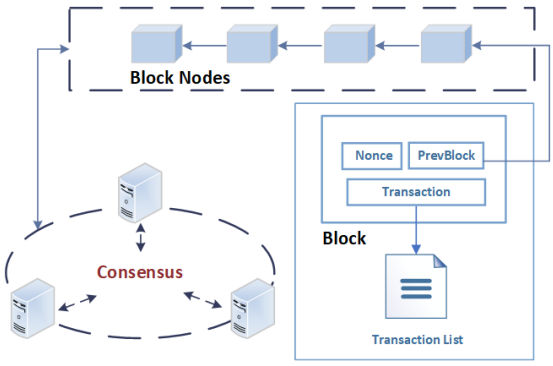


FIGURE 1. Blockchain structure.

1) CONSENSUS MECHANISM

A crucial component of the blockchain is the consensus mechanism, which maintains node consistency and verifies the authenticity and sequence of transactions. The Practical Byzantine Fault Tolerance (PBFT) algorithm was introduced by Castro and Liskov in 1999 as a solution to the Byzantine problem. The PBFT shows a great degree of reliability and robustness in synchronous situations. For PBFT to handle simultaneous Byzantine failures, $N = 3f + 1$ replicas are needed. In light of this, we implement PBFT for IoV. This method works well in the latency-sensitive Internet of Vehicles (IoV) sector and effectively decreases tampering assaults on the central database.

2) SMART CONTRACT

Smart contracts are decentralized blockchain-based programs that can autonomously execute and handle complex contract terms [28]. Nick Szabo first proposed them in 1994. Because they have the benefits of high transparency, non-tampering, and automatic execution, smart contracts are widely employed in a variety of scenarios, including supply chain management, digital identity verification, and digital currency trading.

B. BLS SIGNATURE

Since its inception, the BLS signature scheme has been attracting a lot of attention in the industry for its excellent aggregate signature technology. The scheme not only has excellent key fusion capabilities but also performs efficient m-n multi-signature operations, thus realizing a higher level of security and reliability in distributed systems. The BLS signature scheme allows the master node to integrate signatures from other nodes to form a unified aggregated signature. This design makes the process of verifying information more efficient and convenient in distributed systems. In practical applications, the combined use of the BLS signature algorithm and PBFT algorithm effectively reduces the communication complexity [29]. By reducing the communication cost, the BLS signature scheme provides a strong guarantee for the stable operation of the distributed system.

C. NETWORK MODEL

In this section, the system model of the proposed scheme is introduced, as shown in Figure 2. It mainly consists of two parts, the blockchain network and the vehicle network. In the blockchain network, there are a large number of RSUs and some Edge Servers (ESs), which together constitute the consortium blockchain. In addition, the ES have high computational and storage capacities and are used to manage the real-time synchronization data of the RSUs in their regions. In terms of the vehicle network, it mainly consists of On Board Unit (OBU) registrations. These OBUs are interconnected through wireless communication technology to form a local vehicle network. To support the authentication phase, the smart contract is deployed on the RSU. These smart

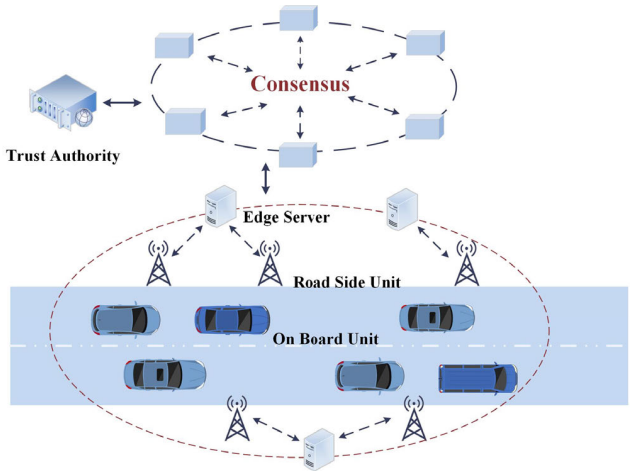


FIGURE 2. Proposed network model.

contracts can automatically perform predefined operations and conditional judgments to provide intelligent support. There are core four entities in the proposed system model.

1) TRUST AUTHORITY (TA)

TA is usually operated by a government department or state-owned enterprise, so it can be considered a fully trusted authority for initializing networks as well as registering vehicles. At the same time, TA can track illegal vehicles. It is generally considered to be quite powerful in terms of data communication and computation.

2) EDGE SERVER (ES)

ES is the regional service manager and is primarily responsible for providing computing and storage resources to the vehicles. They are deployed in the edge network, are usually located near the RSU, and belong to the same edge node as the RSU. The primary responsibility of the ES is to manage all RSUs and validate OBUs in its area. Each ES maintains a public ledger to record the access records of the vehicles and acts as a node to write the write operation authentication results to the general ledger through a consensus algorithm.

3) ROAD SIDE UNIT (RSU)

RSU is an infrastructure located on the side of the road, which is attributed to a specific area and managed by the ES. Due to its limited communication range, it can only communicate wirelessly with vehicles neighboring a specific area based on the Dedicated Short Range Communication (DSRC) protocol. In addition, the RSU contains signaling and receiving modules for data exchange, along with a smart contract module. These RSUs are deployed in edge networks to provide comprehensive coverage for road traffic.

4) ON BOARD UNIT (OBU)

OBU helps passengers and drivers by providing intelligence on road accidents, traffic conditions, driving safety, and terrain services, maintaining valuable information about vehicle equipment, and communicating, calculating, and storing data. The OBU is installed on the vehicle and utilizes wireless communication technology to help the vehicle communicate with other parties, such as vehicles or RSUs.

IV. PROPOSED SCHEME

In this section, the detailed authentication mechanism flow of this scheme will be presented, including five phases, initialization, registration, authentication, consensus, and re-authentication phases. As shown in Figure 3, the last three phases are mainly shown since the initialization and registration phases are executed only once in most cases. In addition, the notation used in this scheme is shown in Table 1.

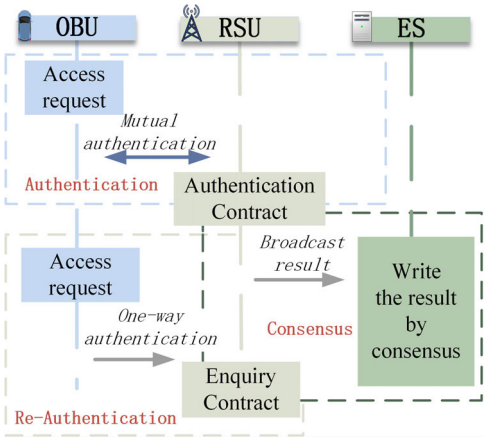


FIGURE 3. Scheme flowchart.

TABLE 1. Notation in proposed scheme.

Symbol	Description
ID_X	Identify of X
PK_X	The public of X
SK_X	The private key of X
$H(\cdot)$	One-way hash function
$Sign_X(B)$	Sign B with X
$\oplus$	XOR operation
$\|\|$	Concatenation operation

A. INITIALIZATION PHASE

To the initialization of the system, TA selects an elliptic curve $E$and sets its public parameters n and P, and then selects a random number $SK_{TA} \in Z_q^*$ as the master private key of the system, generates a public key $PK_{TA} = SK_{TA} \cdot P$, and selects a one-way hash function$H(\cdot)$. The RSU and the ES form a consortium blockchain, and the smart contract is deployed in the RSU. After that, the ES and the RSU generate the private

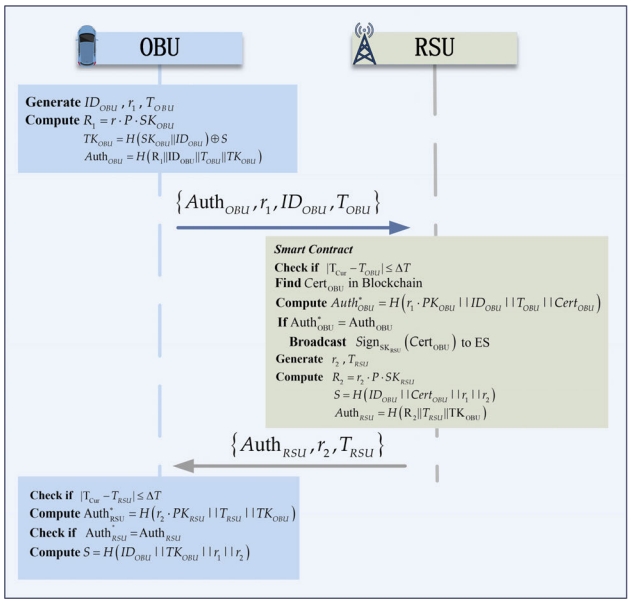


FIGURE 4. Mutual authentication phase.

key, and compute the corresponding public key, based on the parameters of the system that is produced.

B. REGISTRATION PHASE

During this phase, the OBU registers with the TA to join the vehicle network. The OBU registration process is as follows.

1) First, OBU is composed $\mathrm{ID_{OBU}}$ based on a hash of its MAC address. Since the MAC address is unique, it ensures that the real identity of the OBU is unique.

2) Then, OBU generates a timestamp T_{OBU}, and encrypts the message $M_1 = \{ID_{OBU}, T_{OBU}\}$ with $\mathrm{PK_{TA}}$ and send it to TA to request registration.

3) After TA receives message M_1, it first decrypts it and then verifies T_{OBU}. If it passes, it continues the registration request; otherwise, it refuses to the registration request.

4) TA randomly selects an integer $SK_{OBU} \in Z_q^*$, as the private key of OBU, then computes $PK_{OBU} = SK_{OBU} \cdot P$, as the public key of the OBU, and signs $S = Sign_{SK_{RA}}(ID_{OBU})$. Send the message $M_2 = \{SK_{OBU}, PK_{OBU}, S\}$ to OBU, and through the secure channel and store $Cert_{OBU} = H(SK_{OBU}||ID_{OBU}) \oplus H(SK_{TA}||ID_{OBU})$ to the blockchain at the same time to complete the registration.

C. AUTHENTICATION PHASE

After registration, the OBU authenticate by mutual authentication with nearby RSU. After completing the authentication, the authentication result will be sent to ES by RSU through broadcasting, and ES writes it into the blockchain ledger through the consensus algorithm. When OBU initiates access for the first time, the whole operation process is shown in Figure 4, which includes the following steps.

1) OBU is composed $\mathrm{ID_{OBU}}$ based on a hash of its MAC address. Since the mac address is unique, it ensures that the real identity of the OBU is unique.

2) OBU generates a number r_1, timestamp T_{OBU}, computes $R_1 = r_1 \cdot P \cdot SK_{OBU}$, $TK_{OBU} = H(SK_{OBU}||ID_{OBU}) \oplus S$, $Auth_{OBU} = H(R_1||ID_{OBU}||T_{OBU}||TK_{OBU})$, then, encrypts the message $M_3 = \{Auth_{OBU}, r_1, ID_{OBU}, T_{OBU}\}$ with $\mathrm{PK_{RSU}}$ and send it to nearby RSU.

3) RSU receives the message and decrypts it, automatically triggering the authentication contract. It first generates a timestamp T_{Cur}, and checks $|T_{Cur} - T_{OBU}| \le \Delta T$, where ΔT is the maximum legal time interval. If it is not satisfied, the message is discarded, and vice versa, the next step is taken.

4) It queries the corresponding $\mathrm{Cert_{OBU}}$ according to $\mathrm{ID_{OBU}}$, computes $Auth^*_{OBU} = H(r_1 \cdot PK_{OBU}||ID_{OBU}||T_{OBU}||Cert_{OBU})$, verifies $\mathrm{Auth^*_{OBU}} = \mathrm{Auth_{OBU}}$. If the verification fails, the connection is broken.

5) Otherwise, RSU adds signature $Sign_{SK_{RSU}}(Cert_{OBU})$ to the authentication result, and then broadcasts it to the ES. At the same time, RSU generates a random number r_2 and a timestamp T_{RSU}, computes $R_2 = r_2 \cdot P \cdot SK_{RSU}$, $Auth_{RSU} = H(R_2||T_{RSU}||TK_{OBU})$, and session key $S = H(ID_{OBU}||Cert_{OBU}||r_1||r_2)$, encrypts the message $M_4 = \{Auth_{RSU}, r_2, T_{RSU}\}$ with $\mathrm{PK_{OBU}}$, and sends it to OBU.

6) OBU receives the message and decrypts it, first generating a timestamp T_{Cur}, and verifying $|T_{Cur} - T_{RSU}| \le \Delta T$. If it is not satisfied, the message is discarded, and vice versa, computes $\mathrm{Auth^*_{RSU}} = H(r_2 \cdot PK_{RSU}||T_{RSU}||TK_{OBU})$,and then checks $Auth^*_{RSU} \stackrel{?}{=} Auth_{RSU}$.If it passes, the authentication is successful and OBU computes the session key $S = H(ID_{OBU}||TK_{OBU}||r_1||r_2)$.

D. CONSENSUS PHASE

When an ES receives a broadcast authentication result from an RSU, it will check it and securely record the result in the blockchain ledger through a consensus algorithm. However, while PBFT consensus nodes exceeds a certain limit, the bandwidth consumption and consensus delay increase significantly, which makes the PBFT algorithm not directly applicable. To solve this problem, this scheme adopts BLS aggregate signature technology to simplify the PBFT consensus process. Meanwhile, to ensure the security of the system and avoid the master node from doing evil, we introduce the reputation score grouping mechanism. This mechanism will score the reputation of each node and divide the nodes into different groups according to the scoring results.

Specifically, initially all nodes score the same. And are randomly divided into consensus nodes and ordinary nodes. The consensus node can be selected as the master node and participate in the consensus, while the ordinary node only updates the results. Assuming that there are $N/2$ consensus nodes and that the number of consensus nodes must be satisfied, to ensure that the system reaches agreement in all

cases. The ES_x speaker is selected by $x = (h \bmod N) + 1$, where represents the block height, while other peers play as congressmen. The specific steps are as follows.

1) Pre-prepare stage: The speaker sends a message: *pre_prepare* $\langle h, v, d, block, Sig_x \rangle$ to all congressmen. here v stands for view identity, d stands for block's summary, the block is the main content of the whole block, containing all the collected transaction information and the signature for each transaction, and the $\mathrm{Sig_x}$ represents the signature of the speaker.

2) Prepare stage: After all the congressmen receiving pre-prepare messages, they will validate the message and check the legitimacy of the digest, block height, view, and signature, and if the validation is inconsistent, then the score of the speaker will be reduced and terminated at the same time. Otherwise deliver the message: *prepare* $\langle h, v, d, Sig_i \rangle$ to the speaker, $\mathrm{Sig_i}$ is the signature of a congressman.

3) Commit stage: The speaker enters the commit phase after receiving messages from at least $2f + 1$ different congressmen. After verification, the valid signatures received will be Aggregated into one signature via BLS signatures and broadcast message: *commit* $\langle h, v, d, aggregation, node \rangle$. Where aggregation denotes the signature generated after aggregation, and node refers to the list of IDs of all the nodes involved in the aggregated signature. Congressmen receiving this message can verify the validity of the signature using the public key of the node involved in the signature.

4) Reply stage: At this point, congressmen receive the aggregated signature from the Speaker, and after verifying that it passes, it generates a Reply message. When more than $f + 1$ Reply messages are received, the system has reached a consensus on this message.

Meanwhile, the master node will confirm the result and provide it to all the nodes and update the score of each node. For those nodes whose judgments are consistent with the final result, we will give them extra points. Otherwise, score reduction will be done. If there are m nodes in the consensus node set that have a lower score than the common node set, then these m nodes are removed from the consensus set and added to the end of the common list. In addition, the m highest scoring nodes in the common set will be attached to the consensus node set and renumbered. This concludes a consensus process.

E. RE-AUTHENTICATION PHASE

Considering that the vehicle is highly mobile, to avoid repeated authentication, when the vehicle initiates the authentication request again, the authentication becomes a query to the blockchain ledger, said step five re-authentication phase comprises.

1) OBU first generates a timestamp T_{OBU}, then, encrypts the message $M_5 = \{ID_{OBU}, T_{OBU}\}$ with $\mathrm{PK_{RSU}}$ and send it to nearby RSU.

2) RSU receives the message and decrypts it, automatically triggering the authentication contract. It first generates a timestamp T_{Cur}, checks $|T_{Cur} - T_{OBU}| \leq \Delta T$, and then queries the corresponding $\mathrm{Cert_{OBU}}$ according to $\mathrm{ID_{OBU}}$, checking whether it is signed and whether it is in the last block of time T_{tran} has no expiration date.

3) If all are satisfied, the authentication is successful. Otherwise, two-way authentication with the RSU is required. Throughout the process, if the TA receives a report of an illegal vehicle, it will randomly access a public ledger, identify it, inform the illegal vehicle that its public key is invalid, and add it to the revocation list.

V. SAFETY ANALYSIS

In this section, we analyze the security of the proposed scheme. First, we propose a threat model for the existence of the scheme. Second, a formal security validation is performed using the Real-Or-Random (ROR) oracle model and AVISPA tool. Finally, an informal security validation is performed to fully evaluate the security properties of the proposed scheme.

A. ATTACK MODEL

The network model put forward in this paper states that a TA that is backed by a bank, government, or huge enterprise is a completely reliable authority that prevents adversaries from accessing any data that is created or stored. ESs, RSUs, and cars are semi-trusted entities because they are commonly found on the road. While the vehicle authentication step is susceptible to assaults, we presume that the system setup and car registration stages are secure. A network model's confidentiality, integrity, and availability can be compromised in a number of ways, according to the CIA information security model. An attacker can compromise confidentiality, take genuine credentials, and intercept packets from the sender, among other things. Also, an attacker can masquerade as a legitimate entity and modify and forge packets, compromising integrity and availability.

B. FORMAL SECURITY ANALYSIS USING ROR ORACLE MODEL

The ROR oracle model is used to prove that the proposed scheme is secure. The ROR oracle model is a commonly used tool for security proofs of authentication protocols and its validity has been widely recognized. The notation used in this proof is shown in Table 2.

TABLE 2. Notation in proposed scheme.

Symbol	Description
A	Adversary
OBU, RSU, ES	Vehicle, RSU, ES
$\prod OBU^m$	The instance of Vehicle
$\prod RSU^n$	The instance of RSU
$\prod ES^x$	The instance of ES
Y	$Y = \{\prod OBU^m, \prod RSU^n, \prod ES^x\}$

TABLE 3. Comparison of communication cost byzantine consensus algorithms.

Consensus Phase	PBFT	SG-PBFT	Proposed
Pre-prepare	$N-1$	$N/2-1$	$N/2-1$
Prepare	$(N-1)(N-1)$	$(N/2-1)(N/2-1)$	$N/2-1$
Commit	$N(N-1)$	$N/2-1$	$N/2-1$
Total	$2N(N-1)$	$(N/2-1)(N/2+1)$	$3(N/2-1)$

In the ROR oracle model, A can execute the following queries:

1) Execute (Υ):A can eavesdrop on any messages between public channels.

2) Send (Υ, Msg): Similar to a man-in-the-middle attack, Acan send a message Msg to any entity in Υ and get feedback accordingly.

3) Corrupt (Υ): A sends this query to any entity in Υ, gets the returned secret value, and makes it public.

4) Reveal (Υ): A applies the query to reveal a session key between Y entities.

5) Test (Υ): A applies the query by flipping the coin (c, $c \in \{0, 1\}$). If $c = 1$, A can get the correct session key; conversely $c = 0, A$ gets a random value equal to the length of the session key.

Definition 1 (Semantic Security): The corresponding value c^* can be obtained by A by Test (Υ), and if $c^* = c$, then A breaks the semantic security of the proposed scheme. The advantage of A is described as:

$$Adv_A^P = \left|2\Pr\left[c = c^*\right] - 1\right| \tag{1}$$

where c, c^*,a ndPr $\left[c^* = c\right]$, denotes the correct, guessed, and probability that A succeeds, respectively. If there exists a very small positive number ε such that $Adv_A^P \leq \varepsilon$ holds, it indicates that the proposed scheme is safe.

Definition 2 (ECDLP: Elliptic Curve Discrete Logarithm Problem): Given two points Q, P on an elliptic curve E, compute a scalar x such that $Q = x \cdot P$, where $x \cdot P$ denotes the base P point multiplied by x.

Theorem: Assuming A on the proposed scheme in polynomial time t_p, we can define the advantage of attacking and destroying the proposed scheme as:

$$Adv_A^P(t_P) \leq \frac{q_h^2}{\left|hash_f\right|} + 2Adv_A^{ECDLP}(t_P) \tag{2}$$

where q_h, $\left|hash_f\right|$, and Adv_A^{ECDLP},denotes the number of hash queries, the range space of a one-way collision-resistant hash function $H(\cdot)$, and A in polynomial time t_p destroys the advantage of ECDLP.

Proof: The theorem is substantiated through three games to prove the security of t the proposed scheme. Define an event $\mathrm{Succ}_{\mathrm{A}}^{\mathrm{Game_k}}$ that represents the probability that A tries to guess a random bit c of $Game_k$. The advantage of A winning the game is defined as $\Pr[Succ_A^{Game_K}]$, followed by the exposition of each game.

1) Game_0 : This game simulates A attacks in terms of the semantic security of the proposed scheme. A chooses a random bit c, which is available according to the semantic security of Definition 1:

$$Adv_A^P = \left|2\Pr[Succ_A^{Game_0}] - 1\right| \tag{3}$$

2) Game_1 : This game simulates an eavesdropping attack. In the proposed scheme, A uses Execute(Υ) to obtain messages $M_3 = \{\mathrm{Auth}_{OBU}, r_1, ID_{OBU}, T_{OBU}\}, M_4 = \{\mathrm{Auth}_{RSU}, r_2, T_{RSU}\}$, and $M_5 = \{\mathrm{ID}_{OBU}, T_{OBU}\}$ transmitted over the public channel during the authentication phase and tries to obtain the session key. A executes Reveal (Υ) and Test (Υ) to verify the correctness of the recovery key. To get the session key $S = H(ID_{OBU}||TK_{OBU}||r_1||r_2)$, it is necessary to obtain r_1, r_2, ID_{OBU}, and TK_{OBU} at the same time. For r_1, r_2 is randomly generated and fresh each time. For ID_{OBU}, it is composed based on the MAC hash of the OBU, which is unique and tamperproof. and TK_{OBU} is constructed by the hash function $H(\cdot)$.Therefore, A cannot recover the session key in polynomial time t_p, $G\mathrm{ame}_0$ and $G\mathrm{ame}_1$ remain indistinguishable.

$$Pr[\mathrm{Succ}_{\mathrm{A}}^{\mathrm{Game_1}}] = Pr[\mathrm{Succ}_{\mathrm{A}}^{\mathrm{Game_0}}] \tag{4}$$

3) Game_2 : A performs Send (Υ, Msg) and Corrupt (Υ) to launch an active attack. A tries to compute the session key by acquiring r_1, $\mathrm{ID_{OBU}}$, and T_{OBU}, aiming to produce authentication details and procure a response. Nevertheless, regardless of obtaining this information, A cannot generate authentication information. This impossibility arises from the irreversibility and randomness of the hash function, coupled with the necessity to solve the ECDLP problem. Consequently, Game_2 and Game_1 remain indistinguishable. Leveraging the principles of the birthday paradox, the subsequent conclusion is reached:

$$\begin{aligned}&\left|\Pr[Succ_A^{Game_2}] - \Pr[Succ_A^{Game_1}]\right| \\ &\leq \frac{q_{hd}^2}{2\left|hash_d\right|} + Adv_A^{ECDLP}(t_p)\end{aligned} \tag{5}$$

At the end of the game, to guess the correct c, get the following result:

$$\Pr[Succ_A^{Game_2}] = \frac{1}{2} \tag{6}$$

It can be deduced from Equation. (1) (6):

$$\frac{1}{2}Adv_A^P(t_p) = \left|\Pr[Succ_A^{Game_0}] - \frac{1}{2}\right| \tag{7}$$

According to equations (3), (4), (5) and the trigonometric inequality, one can get:

$$\begin{aligned}\frac{1}{2}Adv_A^P(t_p) &= \left|\Pr[Succ_A^{Game_0}] - \Pr[Succ_A^{Game_2}]\right| \\ &= \left|\Pr[Succ_A^{Game_1}] - \Pr[Succ_A^{Game_2}]\right|\end{aligned}$$

$$\leq \frac{q_h^2}{2\left|hash_f\right|} + Adv_A^{ECDLP}(t_P) \tag{8}$$

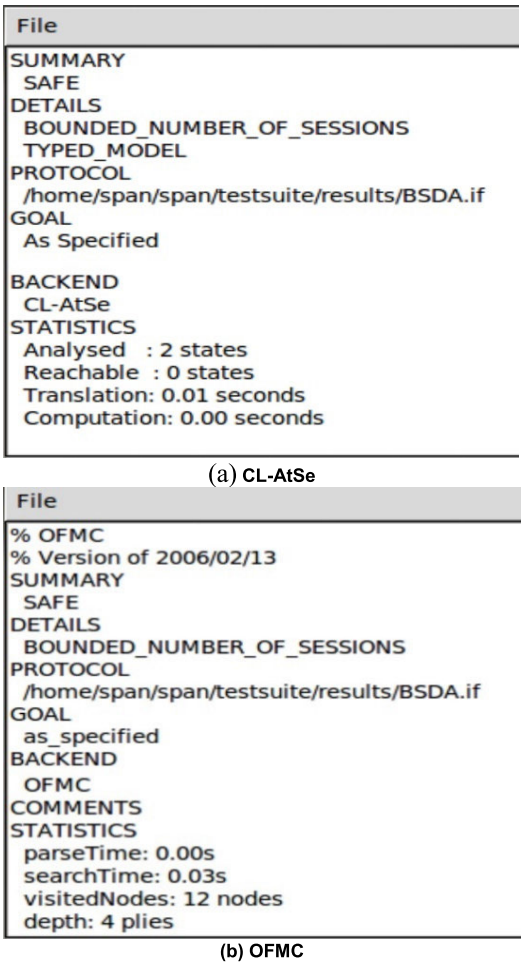
Hence, we can get the final result.

$$Adv_A^P(t_P) \leq \frac{q_h^2}{\left|hash_f\right|} + 2Adv_A^{ECDLP}(t_P) \tag{9}$$

C. FORMAL SECURITY VERIFICATION USING AVISPA TOOL

To fully validate the security of the proposed scheme, we use Automated Validation of Internet Security Protocols and Applications (AVISPA) to provide formal security validation. The AVISPA toolkit is integrated into a virtual machine called SPAN, an open-source application suite that supports analyzing and validating security protocols. The AVISPA integrates four backends, namely CL-AtSe, SATMC, OFMC, and TA4SP. AVISPA uses a formal language, HLPSL, which is well suited for applications in IoT scenarios, including protocol verification under the Internet of Vehicles.

In this scheme, the roles involved are OBU, RSU, ES, Session and Environment. Generally, if the proposed scheme is secure on OFMC and CL-AtSe models, then the scheme is resistant to retransmission attacks and man-in-the-middle attacks. After executing the code on AVISPA, the relevant results are shown in Figure 5. It can be seen that the scheme is proved to be safe on both OFMC and CL-AtSe models.



(a) CL-AtSe

(b) OFMC

FIGURE 5. Simulation results of the proposed scheme.

D. INFORMAL SECURITY ANALYSIS

1) SYSBIL ATTACK

In this scheme, each node has a unique ID in the network, which consists of a hash of MAC which is unique and cannot be modified. In addition, mutual authentication is performed when the entity is accessed for the first time, and the result of the access is recorded in the blockchain ledger, which is traceable and unforgeable. Therefore, Sybil attackers cannot forge legitimate nodes in the network to communicate with other nodes.

2) SPOOFING ATTACK

Since all the nodes in the IoV must be registered first if the nodes interact with each other, they must be verified through authentication, and each node must verify the identification it holds to prove its unique identity so the attacker can not disguise as the identity of other illegal nodes to carry out spoofing attacks.

3) FORGERY ATTACK

To successfully forge the identity of a node, the attacker has to compute the private key and public parameters generated by ECC, which is computationally very difficult. Hence, it is difficult for the attacker to forge the node's identity successfully.

4) MAN IN THE MIDDLE ATTACK

An attacker can intercept and alter the messages transmitted between the vehicle and the RSU. However, the intermediate messages transmitted can only be accessed with the private keys of both parties. All messages transmitted over the channel contain timestamp parameters to verify the reasonableness of the time when the message was sent, and, in a man-in-the-middle attack, the attacker has to forge legitimate identities, and the forgery attack is proven preventable.

5) DISTRIBUTED DENIAL SERVICE ATTACK

Authentication results are added to the blockchain by optimizing PBFT, and the coalition nodes will identify the malicious node if the attacker prevents the authentication message from being sent and uses a false signature. After the consensus is over, the malicious node can no longer participate. In addition, this mechanism is built based on blockchain with decentralized characteristics. The authentication results are stored in the distributed ledger and updated in real time, and node failures do not affect other nodes' data, effectively resisting DDoS attacks.

6) REPLAY ATTACK

The scheme proposed in this paper uses timestamps, which act as a dynamic and unrepeatable element in each

authentication process, making each request unique and thus effectively resisting replay attacks.

VI. PERFORMANCE EVALUATION

In this section, we first analyze the proposed optimized consensus algorithm to reveal its intrinsic mechanisms and advantages, and then comprehensively evaluate the computational and communication costs of the scheme. Finally, the scheme is simulated using SUMO simulation to verify the practicality and effectiveness of the scheme.

A. CONSENSUS EVALUATION

The traditional PBFT consensus algorithm consumes many communication resources to evaluate our scheme's consensus performance fully. We assume that there is a total of N nodes in the IoV, and all nodes are required to participate in the consensus in PBFT. At the same time, the consumption of communication resources mainly originates from the three phases of pre-preparation, preparation, and submission. Therefore, the number of communications in a consensus process of the PBFT consensus algorithm is $2N\ (N-1)$. For SGPBFT [26] divides the nodes equally into consensus nodes and candidate nodes, and the consensus nodes participate in the consensus process. Thus,$(N/2-1)\ (N/2+1)$ times are required to complete the consensus. As for our scheme, because of the intro-duction of BLS aggregation signatures, the broadcast in the preparation phase becomes unicast to the master node, and a packet scoring mechanism is added to prevent de-pendency on the master node. Therefore, the total number of communications is $3\,(N/2+1)$. able III shows the, we list the number of communications for PBFT, SGPBFT, and each stage of our scheme. It's obvious that both PBFT and SG-PBFT communication complexities is $O\left(N^2\right)$, and our scheme is only $O\,(N)$.

For this scheme, BLS aggregated signatures are introduced, and we use the library Crypto++ in C++ for evaluation. The experimental results are shown in Figure 6, where the time consumed gradually increases from 50 aggregated signatures to 500 aggregated signatures. However, the figure 6 shows that when the number of aggregate signatures reaches 500, the time consumed is only about 0.12 ms, so introducing BLS aggregate signatures reduces the cost of the overall PBFT consensus algorithm.

B. AUTHENTICATION EVALUATION

1) COMPUTATION COST

To analyze the computational cost, we use the total execution time of all cryptographic primitives in the protocol as the basis for comparing the computational cost, and calculate the time cost associated with hash functions, elliptic curve multiplication, elliptic curve addition, dissimilarity time, and so on, and the time required for each operation is T_{ha}, T_{emc}, T_{eac}, and T_{XOR}, respectively. Researchers and cryptographers commonly rely on the Multiprecision Integer and Rational Arithmetic Library (MIRACL) [30] for testing the execution

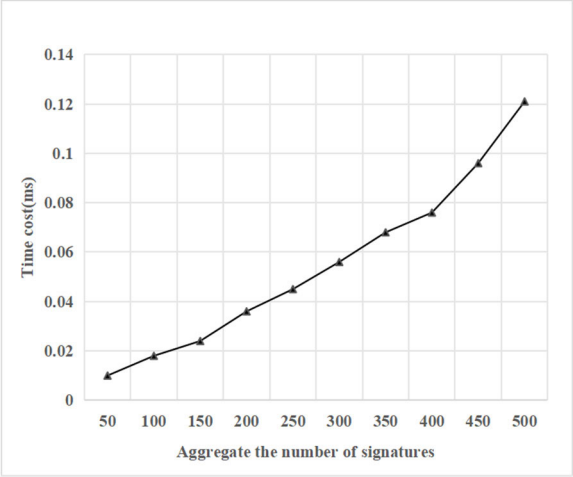


FIGURE 6. Time consumed for aggregating n signatures.

TABLE 4. Approximate runtime required for cryptographic primitive.

Operation	Description	Time(ms)
T_{ha}	One-way hash function	0.055
T_{ecm}	Point multiplication operation related to the ECC	0.674
T_{eca}	Point addition operation related to the ECC	0.002
T_{CKDf}	Cryptographic key derivation function	0.674

time of various cryptographic primitives. The library provides a unified platform for cryptographic experiments, allowing the performance of various cryptographic primitives to be compared and measured under the same conditions. In this paper, we summarize the approximate running times of different cryptographic primitives running in MIRACL based on the experimental results of [31] and [32], as shown in Table 4.

The computational cost of the proposed scheme mainly originates from its registration and authentication phases. In the registration phase, the OBU needs to submit a registration application to the TA to complete its process of joining the vehicular network. Entering the authentication phase, OBU and RSU must perform a mutual authentication process, which is to ensure the authenticity and legitimacy of each other's identity. It is worth mentioning that in the re-authentication phase, only simple query operations need to be performed as the blockchain technology has already recorded the previous authentication results, thus significantly reducing the additional computational cost. Therefore, the total computational cost of the proposed scheme is $8T_{ha}+\ 6T_{ecm}$. To further validate the advantage of the proposed scheme in computational cost, we compared with its related schemes with [33], [34], [35], and [36] as shown in Table 5. The proposed scheme outperforms all the other schemes.

TABLE 5. Comparison of computational cost.

scheme	Computation cost	Time(ms)
[33]	$36\,T_{ha}+5\,T_{ecm}+T_{eca}$	5.325
[34]	$18\,T_{ha}+10\,T_{ecm}+4\,T_{CKDf}$	10.426
[35]	$10\,T_{ha}+10\,T_{ecm}+12\,T_{eca}$	7.284
[36]	$21\,T_{ha}+12\,T_{ecm}$	9.243
Proposed	$8\,T_{ha}+6\,T_{ecm}$	4.484

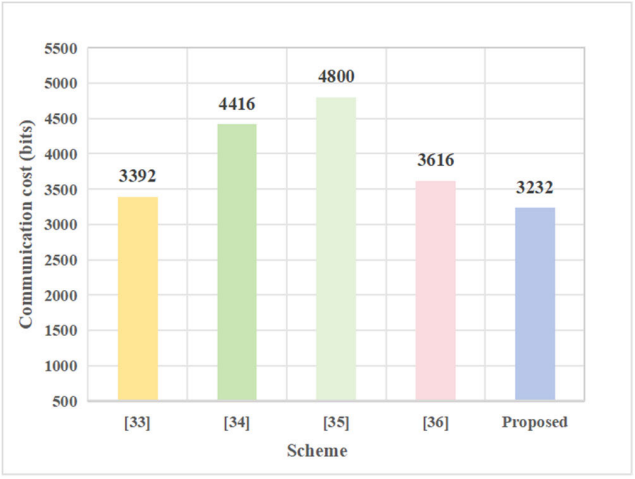


FIGURE 7. Comparison of communication cost.

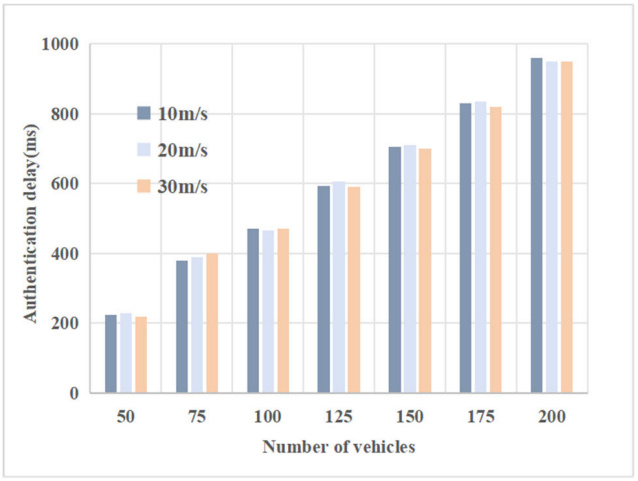


FIGURE 8. Comparison of authentication delay for different speeds.

TABLE 6. Simulation parameters.

Parameter	Value
Simulation area	1000m×1000m
Simulation time	30s
Number of vehicles	50-200
Speed of vehicle	10-30m/s
Wireless protocol	802.11p
Channel bandwidth	6Mbps

C. COMMUNICATION COST

For the communication cost, the number of bits of all messages exchanged during the communication of the authentication scheme is used as the communication cost of the scheme. We consider the identity, random number, timestamp, hash function, and elliptic curve cryptography (ECC) as 160 bits, 160 bits, 32 bits, 256 bits, and 160 bits, respectively. We similarly compare the proposed scheme with [33], [34], [35], and [36]. As shown in Figure 7. In the scheme of Wazid et al. [33] the number of messages exchanged is 3, which are 1024 bits, 1344 bits, and 1024 bits, respectively. Therefore, the total communication cost of this scheme is 3392 bits. In contrast, the total communication cost of the scheme Eddine et al. [34] and Zhang et al. [35] reaches 4416 bits and 4800 bits, respectively. For the scheme Dwivedi et al. [36] which requires the exchange of 4 messages, the total communication cost is 3616 bits.

In this scheme, the communication cost mainly comes from the registration, authentication, and re-authentication phases. Firstly, in the registration phase, $M_1 = \{ID_{OBU}, T_{OBU}\}$, and $M_2 = \{SK_{OBU}, PK_{OBU}, S\}$ need to be exchanged, hence the communication cost is (352+736) =1088 bits. Next, in the authentication phase, a mutual authentication process is required since this is the first access between the OBU and the RSU. This means that both parties need to send and receive authentication messages$M_3 = \{\text{Auth}_{OBU}, r_1, ID_{OBU}, T_{OBU}\}$, and $M_4 = \{\text{Auth}_{RSU}, r_2, T_{RSU}\}$ to each other to ensure the authenticity of the identity and security of the communication. Therefore, the communication cost in this phase is (1184+608) = 1792 bits. In the re-authentication phase, the communication cost is 352 bits because the authentication result has been recorded in the blockchain. Therefore, the total communication cost required for exchanging the messages in this scheme is (1088+1792+352) = 3232 bits. Compared with the other schemes, this scheme shows a significant advantage in communication cost shows a significant advantage.

1) AUTHENTICATION DELAY

Our scheme is simulated using SUMO, an open source traffic simulation software developed in C++, to measure the average authentication delay. We define the authentication delay as the time consumed by the vehicle to complete the whole authentication process. The parameters of the experimental simulation are shown in Table 6. As shown in Figure 8, we set the vehicle speed to 10m/s, 20m/s, 30m/s, respectively, and we can see that the total authentication delay gradually increases with the increase of the vehicle. Meanwhile for different speeds, the total authentication delay does not change significantly.

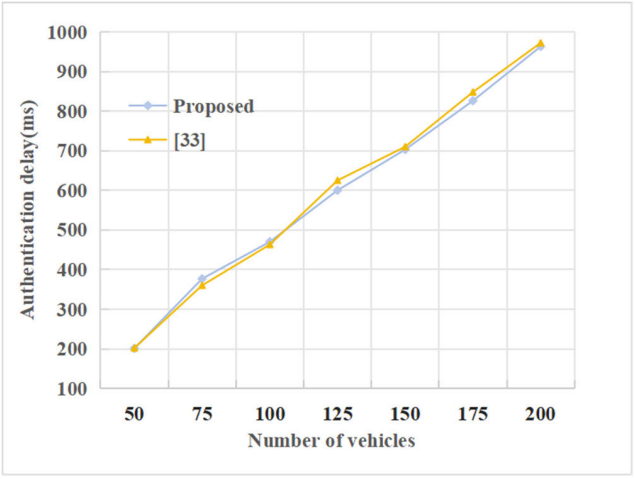


FIGURE 9. Average authentication delay.

In the computation and communication cost analysis, according to Table 5 and Figure 7, it can be seen that the scheme of Wazid et al. [33] is lower than other schemes, so in this paper, the scheme of Wazid et al. [33] is used as a benchmark to compare with the average authentication delay of this scheme. The simulation results are shown in Figure 9, and it is obvious that the results of this scheme and the scheme of Wazid et al. [33] are similar. Therefore, the average authentication delay of the proposed scheme is in a reasonable range and has the value and feasibility of practical application.

VII. CONCLUSION AND FUTURE WORK

In this paper, we propose a blockchain-based secure distributed authentication scheme for Telematics, which solves many problems of traditional Telematics access authentication schemes. Smart contracts and blockchain technology realize automation and a more efficient and transparent authentication process in the IoV authentication process. The optimized PBFT consensus algorithm securely adds the authentication results to the blockchain, avoiding duplicate authentication. The ROR oracle model and AVISPA tool comprehensively evaluate the security of this scheme. In addition, a detailed performance evaluation shows that the scheme excels in computation and communication. With the help of SUMO simulation, the experimental results show that the scheme has the value and feasibility of practical application. In the future, we will further investigate how to optimize this scheme to adapt to the rapid development and changes of IoV. At the same time, we will also explore how to apply this scheme to large-scale scenarios and other mobile areas of the Internet of Things in real life, to provide secure and efficient communication solutions for more scenarios.

REFERENCES

[1] J. Contreras-Castillo, S. Zeadally, and J. A. Guerrero-Ibañez, "Internet of Vehicles: Architecture, protocols, and security," *IEEE Internet Things J.*, vol. 5, no. 5, pp. 3701–3709, Oct. 2018.

[2] I. Makhdoom, M. Abolhasan, J. Lipman, R. P. Liu, and W. Ni, "Anatomy of threats to the Internet of Things," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 2, pp. 1636–1675, 2nd Quart., 2019.

[3] M. Dutta and J. Granjal, "Towards a secure Internet of Things: A comprehensive study of second line defense mechanisms," *IEEE Access*, vol. 8, pp. 127272–127312, 2020.

[4] M. Serror, S. Hack, M. Henze, M. Schuba, and K. Wehrle, "Challenges and opportunities in securing the industrial Internet of Things," *IEEE Trans. Ind. Informat.*, vol. 17, no. 5, pp. 2985–2996, May 2021.

[5] I. Ali, S. Sabir, and Z. Ullah, "Internet of Things security, device authentication and access control: A review," 2019, *arXiv:1901.07309*.

[6] J. Heo, B. Kang, J. M. Yang, J. Paek, and S. Bahk, "Performance-cost tradeoff of using mobile roadside units for V2X communication," *IEEE Trans. Veh. Technol.*, vol. 68, no. 9, pp. 9049–9059, Sep. 2019.

[7] L. Mendiboure, M. A. Chalouf, and F. Krief, "Survey on blockchain-based applications in Internet of Vehicles," *Comput. Electr. Eng.*, vol. 84, Jun. 2020, Art. no. 106646.

[8] W. Zhou, Y. Jia, A. Peng, Y. Zhang, and P. Liu, "The effect of IoT new features on security and privacy: New threats, existing solutions, and challenges yet to be solved," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 1606–1616, Apr. 2019.

[9] B. W. Nyamtiga, J. C. S. Sicato, S. Rathore, Y. Sung, and J. H. Park, "Blockchain-based secure storage management with edge computing for IoT," *Electronics*, vol. 8, no. 8, p. 828, Jul. 2019.

[10] M. Jiang and X. Qin, "Distributed ledger technologies in vehicular mobile edge computing: A survey," *Complex Intell. Syst.*, vol. 8, no. 5, pp. 4403–4419, Oct. 2022.

[11] J. Grover, "Security of vehicular ad hoc networks using blockchain: A comprehensive review," *Veh. Commun.*, vol. 34, Apr. 2022, Art. no. 100458.

[12] D. Zheng, C. Jing, R. Guo, S. Gao, and L. Wang, "A traceable blockchain-based access authentication system with privacy preservation in VANETs," *IEEE Access*, vol. 7, pp. 117716–117726, 2019.

[13] J. Noh, S. Jeon, and S. Cho, "Distributed blockchain-based message authentication scheme for connected vehicles," *Electronics*, vol. 9, no. 1, p. 74, Jan. 2020.

[14] A. Vangala, A. K. Das, A. Mitra, S. K. Das, and Y. Park, "Blockchain-enabled authenticated key agreement scheme for mobile vehicles-assisted precision agricultural IoT networks," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 904–919, 2023.

[15] K. Xue, X. Luo, Y. Ma, J. Li, J. Liu, and D. S. L. Wei, "A distributed authentication scheme based on smart contract for roaming service in mobile vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 5, pp. 5284–5297, May 2022.

[16] H. Xue, D. Chen, N. Zhang, H.-N. Dai, and K. Yu, "Integration of blockchain and edge computing in Internet of Things: A survey," *Future Gener. Comput. Syst.*, vol. 144, pp. 307–326, Jul. 2023.

[17] B. Hildebrand, M. Baza, T. Salman, S. Tabassum, B. Konatham, F. Amsaad, and A. Razaque, "A comprehensive review on blockchains for Internet of Vehicles: Challenges and directions," *Comput. Sci. Rev.*, vol. 48, May 2023, Art. no. 100547.

[18] R. Yang, F. R. Yu, P. Si, Z. Yang, and Y. Zhang, "Integrated blockchain and edge computing systems: A survey, some research issues and challenges," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 2, pp. 1508–1532, 2nd Quart., 2019.

[19] M. Shen, H. Lu, F. Wang, H. Liu, and L. Zhu, "Secure and efficient blockchain-assisted authentication for edge-integrated Internet-of-Vehicles," *IEEE Trans. Veh. Technol.*, vol. 71, no. 11, pp. 12250–12263, Nov. 2022.

[20] S. Guo, X. Hu, Z. Zhou, X. Wang, F. Qi, and L. Gao, "Trust access authentication in vehicular network based on blockchain," *China Commun.*, vol. 16, no. 6, pp. 18–30, Jun. 2019.

[21] Y. Yao, X. Chang, J. Misic, V. B. Misic, and L. Li, "BLA: Blockchain-assisted lightweight anonymous authentication for distributed vehicular fog services," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 3775–3784, Apr. 2019.

[22] A. Wu, Y. Guo, and Y. Guo, "A decentralized lightweight blockchain-based authentication mechanism for Internet of vehicles," *Peer-Peer Netw. Appl.*, vol. 16, no. 3, pp. 1340–1353, May 2023.

[23] S.-K. Kim, "The trailer of strategic alliance for blockchain governance game," 2019, *arXiv:1903.11172*.

[24] S. Tang, Z. Wang, J. Jiang, S. Ge, and G. Tan, "Improved PBFT algorithm for high-frequency trading scenarios of alliance blockchain," *Sci. Rep.*, vol. 12, no. 1, p. 4426, Mar. 2022.
[25] A. Kumar, L. Vishwakarma, and D. Das, "R-PBFT: A secure and intelligent consensus algorithm for Internet of Vehicles," *Veh. Commun.*, vol. 41, Jun. 2023, Art. no. 100609.
[26] G. Xu, H. Bai, J. Xing, T. Luo, N. N. Xiong, X. Cheng, S. Liu, and X. Zheng, "SG-PBFT: A secure and highly efficient distributed blockchain PBFT consensus algorithm for intelligent Internet of Vehicles," *J. Parallel Distrib. Comput.*, vol. 164, pp. 1–11, Jun. 2022.
[27] Y. Zhou, L. Yu, Z. Jiang, Z. Zhi, J. Kang, and Z. Han, "An improved spectrum trading design based on dynamic credit aggregate-signature blockchain," *IEEE Wireless Commun. Lett.*, vol. 12, no. 4, pp. 625–629, Apr. 2023.
[28] K. Christidis and M. Devetsikiotis, "Blockchains and smart contracts for the Internet of Things," *IEEE Access*, vol. 4, pp. 2292–2303, 2016.
[29] J. Chen, X. Xian, Z. Yang, and W. Liu, "Improved practical Byzantine fault tolerant consensus algorithm combined with BLS aggregating signature," *Appl. Res. Comput.*, vol. 38, no. 7, pp. 1952–1955, 2021.
[30] M. C. Sdk. *MIRACL Cryptographic SDK: Multiprecision Integer and Rational Arithmetic Cryptographic Library*. Accessed: Jun. 2022. [Online]. Available: https://github.com/miracle/MIRACL
[31] S. Roy, S. Nandi, R. Maheshwari, S. Shetty, A. K. Das, and P. Lorenz, "Blockchain-based efficient access control with handover policy in IoV-enabled intelligent transportation system," *IEEE Trans. Veh. Technol.*, vol. 73, no. 3, pp. 3009–3024, Mar. 2024.
[32] S. Saha, B. Bera, A. K. Das, N. Kumar, S. H. Islam, and Y. Park, "Private blockchain envisioned access control system for securing industrial IoT-based pervasive edge computing," *IEEE Access*, vol. 11, pp. 130206–130229, 2023.
[33] M. Wazid, P. Bagga, A. K. Das, S. Shetty, J. J. P. C. Rodrigues, and Y. Park, "AKM-IoV: Authenticated key management protocol in fog computing-based Internet of Vehicles deployment," *IEEE Internet Things J.*, vol. 6, no. 5, pp. 8804–8817, Oct. 2019.
[34] M. S. Eddine, M. A. Ferrag, O. Friha, and L. Maglaras, "EASBF: An efficient authentication scheme over blockchain for fog computing-enabled Internet of Vehicles," *J. Inf. Secur. Appl.*, vol. 59, Jun. 2021, Art. no. 102802.
[35] H. Zhang, Y. Lai, and Y. Chen, "Authentication methods for Internet of Vehicles based on trusted connection architecture," *Simul. Model. Pract. Theory*, vol. 122, Jan. 2023, Art. no. 102681.
[36] S. K. Dwivedi, R. Amin, S. Vollala, and M. K. Khan, "B-HAS: Blockchain-assisted efficient handover authentication and secure communication protocol in VANETs," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 6, pp. 3491–3505, Nov. 2023.



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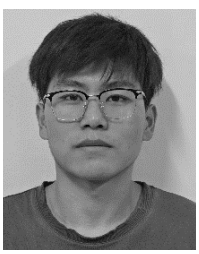
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